

Title for your proposal

1 Driving Motivation

Commercial aviation plays a vital role in regional and international transportation. However, air travel is known to be a major source of greenhouse gases, estimated at around 2.4% of global CO₂ fossil fuel emissions in 2018 [1]. Furthermore, the International Civil Aviation Organization projects these emissions to nearly double by 2050 [2].

Electric aircraft are a potential solution to the challenges of sustainable transportation, with reduced carbon emissions, lower noise levels, and improved mechanical efficiency. Battery lifespan has emerged as a key economic factor [3]. However, fast charge cycles are required to maintain availability [4], which, combined with the high power demand in flight, leads stacks to rapidly deteriorate [5].

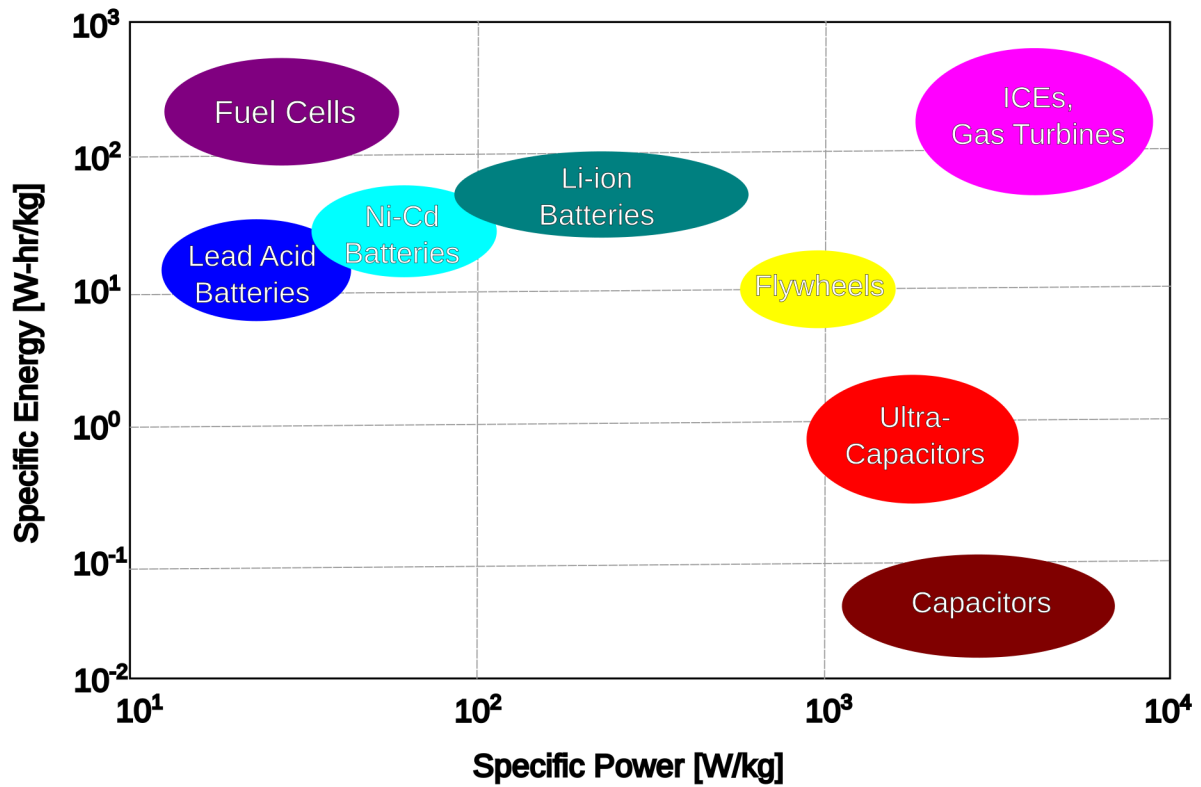


Fig. 1: Specific weight and power of a variety of energy storage devices. No single system is comparable to a traditional ICE, requiring a combination of systems in a hybrid configuration and an EMS.

Hydrogen fuel cell/battery hybrid architectures have been proposed as an alternative to all-electric systems in land-based transportation [6, 7]. In principle, these architectures are easily ported to aircraft. However, even in hybrid configurations batteries represent a major maintenance bottleneck, with lifespans ranging from a few hundred flight hours in a fuel cell/battery set up [8] to less than 2000 hours for a turbine/battery hybrid [9], compared to the over 4000 operating hours [10] in a fuel cell life cycle.

The energy management system (EMS) is key to the performance of hybrid systems as it balances demand across each of the separate power supplies efficiently and safely. Doing so allows the aircraft to leverage the relative advantages of each power source while minimizing weight and extending battery life. The object of the present proposal is the development of an EMS library written for the Modelica language, using the standard multi-physics libraries provided by OpenModelica [11]. In addition to a detailed case study quantifying the impact of route planning, recharge schedules, and flight abnormalities on battery life, the open-source framework will support further research in EMS design and battery prognostics for electrified aviation.

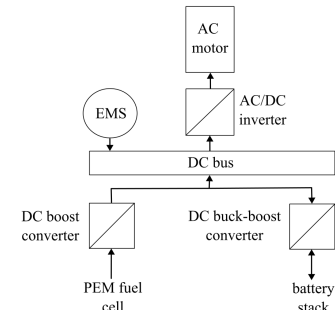
2 Prior Work

2.1 All-Electric Hybrid Architectures

The typically proposed architecture for a hydrogen fuel cell/battery hybrid vehicle is depicted in Fig. 2b. In the most basic application, the fuel cell acts as a simple range extender for the battery [12]; as the battery becomes depleted, the fuel cell ramps up to provide power to the vehicle's electric motor. More usually, the fuel cell acts as the primary power supply for the vehicle, with the battery providing supplemental power. The degree to which the battery is utilized depends upon the application and control scheme. For example, Bauman [7] proposed relying on the battery during transients and at times when the fuel cell would otherwise be operating at a low efficiency. Alternatively, the battery may supply an arbitrary percentage of the load during normal operation [6], or supply load only when demand exceeds what the fuel cell can provide [13]. The concept can further be extended by the addition of an ultracapacitor bank [7, 14, 15]. In general, the role of the ultracapacitors is limited to handling high power transients, such as those experienced during acceleration.



(a) The Cessna 208.



(b) Generalized layout of a proposed power train for a fuel cell-battery hybrid ground vehicle.

Fig. 2: The Cessna 208 and proposed hybrid power train.

Historically the overwhelming majority of effort in developing hybrid electric powertrains, including every paper referenced above, has been dedicated to land vehicles. However, in recent years interest has been growing in applying these methods to the aviation industry. Most efforts at hybridization, such as the 9-seater Electra EL9 Ultra Short [16] and Ampaire Eco Caravan [17] aimed at regional transportation, combine a traditional turbogenerator with variously sized battery stacks. Similarly, in 2023 ZeroAvia adapted the 19-seater Dornier 228 into a turbogenerator/fuel cell hybrid [18].

In comparison, most proposed all-electric aircraft are generally small 1-2 seat craft [19]. For example, the Pipistrel Velis Electro [20] is a battery-powered trainer craft. This trend is driven primarily by the current limitations of battery technology [19]. Their limited range lower power demand reduce the size and weight of the required battery. Furthermore, the low-efficiency of an internal combustion engine relative to a turbogenerator means there is a greater advantage in switching to an all-electric system in small platforms. The prime example of an all-electric hybrid was the X-57 Maxwell developed by NASA [21, 22], with a power train featuring a solid oxide fuel cell, a battery stack, and two inboard electric motors.

2.2 Energy Management System Strategies

Regardless of the exact configuration of the power supply, an EMS is required to coordinate each of the components. The simplest and most common approach is operational mode-based, or rule-based, control [6, 7, 13, 14, 23–25]. In such a scheme, the power supplied by each component is defined in reference to the demand on the overall system [23]. For example, the controller may compare the demand to a reference power for the fuel cell. If demand is less than the reference value, the fuel cell supplies the entire load; if it is greater, the battery stack makes up the difference. This basic structure can then be augmented with additional rules, such as a requirement to maintain the battery's state of charge (SOC) within a given range or a limitation on hydrogen consumption [13, 25].

The three primary alternatives to a rule-based control structure are equivalent consumption minimization strategies (ECMS)[23, 26–28], fuzzy logic based controls [23, 29–31], and predictive controls [23, 32–34]. In equivalent consumption strategies, the object is to minimize the overall hydrogen consumption of the power train, defined by summing the consumption rate of the fuel cell by the equivalent consumption of the other power supplies. In the case of the battery stack, a penalty can be applied so as to maintain the stack's SOC at a given target. Because the consumption rates are dependent on the power supplied by their components, minimizing the overall hydrogen consumption also optimizes the load shared between the fuel cell and battery.

A fuzzy logic controller is a simplistic machine learning method in which an arbitrary number of inputs, typically including load demand and battery SOC, are mapped onto a number outputs using a set of membership functions and logical rules. For example, the SOC might be classed as "Low," "Medium," or "High" by a membership function similar to Fig. 3. After determining the membership of each input according to its function, the results are fed into a table of logical if/then statements to assign an output; e.g., if load demand is "High" and SOC is "Low", the load placed on the fuel cell might be "High." Once the output's membership class has thus been determined, it is defuzzied into a command usable by the system, such as a discreet value for the fuel cell power.

Finally, predictive controls operate by attempting to govern the controller output by forward prediction of the system state. Prediction is performed by a linear time-invariant plant model and projected to a given prediction horizon. At each sampling instant, the system state is measured and compared to the current and future predictions. The controller then optimizes a set of changes in the manipulated variables so as to bring the outputs into agreement with a given setpoint. So, for example [32], a hybrid fuel cell/battery controller might output a load demand, battery SOC, and hydrogen consumption rate against reference values for the demand and SOC. At every time step, the outputs are measured against their predicted output and setpoint. The controller supplies a new prediction and manipulates the fuel cell and battery loads to bring the outputs into conformity, and the process repeats.

All of the above methods have been applied to fuel cell/battery hybrid aircraft. Wise [35] developed a rule-based and ECMS control scheme for comparison with a dynamic programming method, an approach aimed at optimizing the fuel cell power. They found that while the ECMS produced the lowest cost per mission, the dynamic programming approach reduced overall lifetime costs by maximizing battery life, with the rule-based method providing competitive performance. Likewise, in a study by Mojapon [36], a mode-based, ECMS, and classic PI controller, as well as two separate fuzzy logic controllers, were built

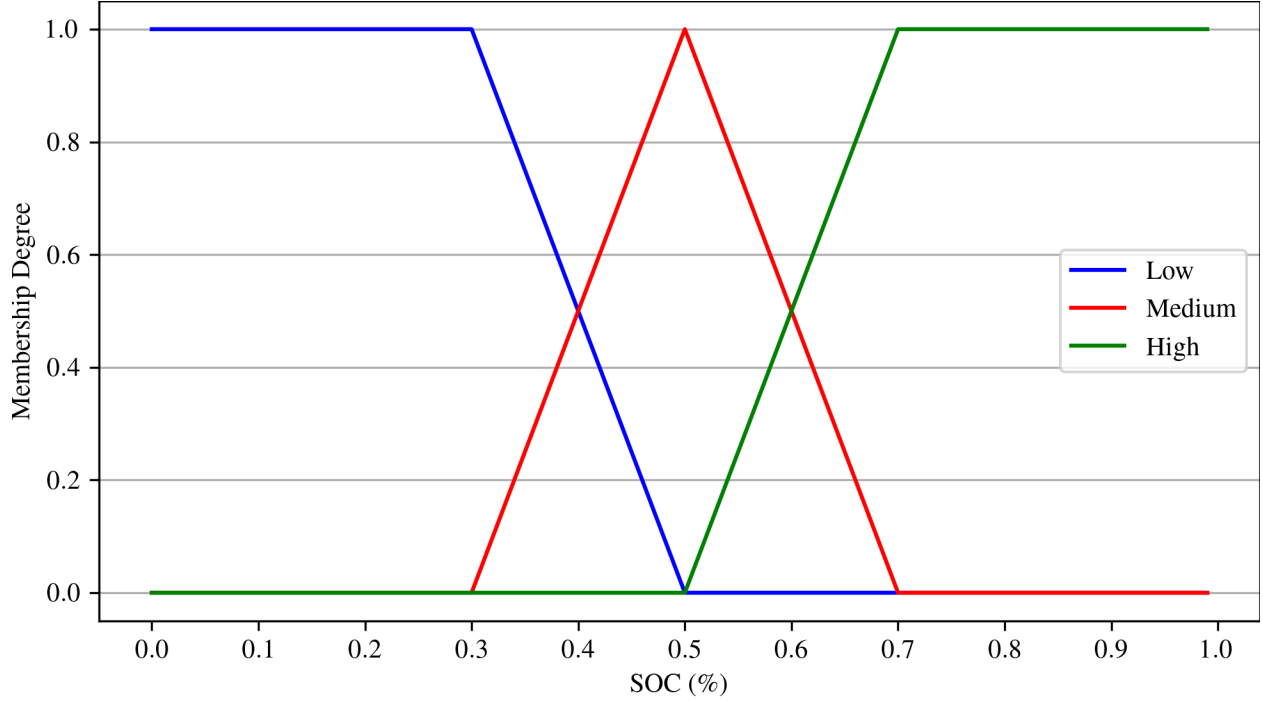


Fig. 3: Generic SOC membership function in a fuzzy logic controller.

and compared for a fuel cell/battery/supercapacity emergency power system. In general, it was found that the mode-based minimized battery and supercapacity stresses, while the fuzzy logic controls minimized the fuel cell stress. However, this study was limited in scale and only applied to aborted landings, rather than a more typical takeoff and cruise. Finally, Li [37] augmented a predictive controller by using fuzzy logic membership functions to regulate a gain function governing the output current of the fuel cell. This method was found to greatly reduce hydrogen consumption and stress on the fuel cell at the expense of the battery.

Generally speaking, most studies of electric hybrid EMSs are purely simulation based. Very few examples of demonstration on hardware exist, and these are generally limited to small drones and UAVs [38, 39]. The platform of choice for these simulations is MATLAB/Simulink, a commonly used commercial multi-physics and design package. For example, the University of Illinois' Center for High-Efficiency Electrical Technologies for Aircraft (CHEETA) [40] has greatly extended the basic models provided with the program, including packages for structural and thermal optimization, cryogenic energy systems, power systems integration, among others; however, usage of these robust tools is still reliant upon purchase of a Simulink license.

2.3 Power Train Modeling and Passive Control

The object of the initial work in this area was the development of a basis for an EMS library written for the Modelica language, using the standard multi-physics libraries provided by OpenModelica [11]. A simplified model of a hybrid fuel cell/battery power supply, as pictured in Fig. 4, was developed from the literature and experimental battery characterization data. The "fc" block consists of a combination of theoretical and empirical relations necessary for describing the behaviour of a Ballard Mk5 fuel cell, supplied with fuel by a liquid hydrogen "tank" block. Cooling for the fuel cell is provided by a closed cooling loop consisting of the fuel cell cooling plate, air-coolant heat exchanger, and hydrogen preheat. The "stack" is a modified version of the OpenModelica standard library's model, incorporating a degradation model and characterized by experimental data. Finally, the "cycle" block combines the power profile generator and load sharing

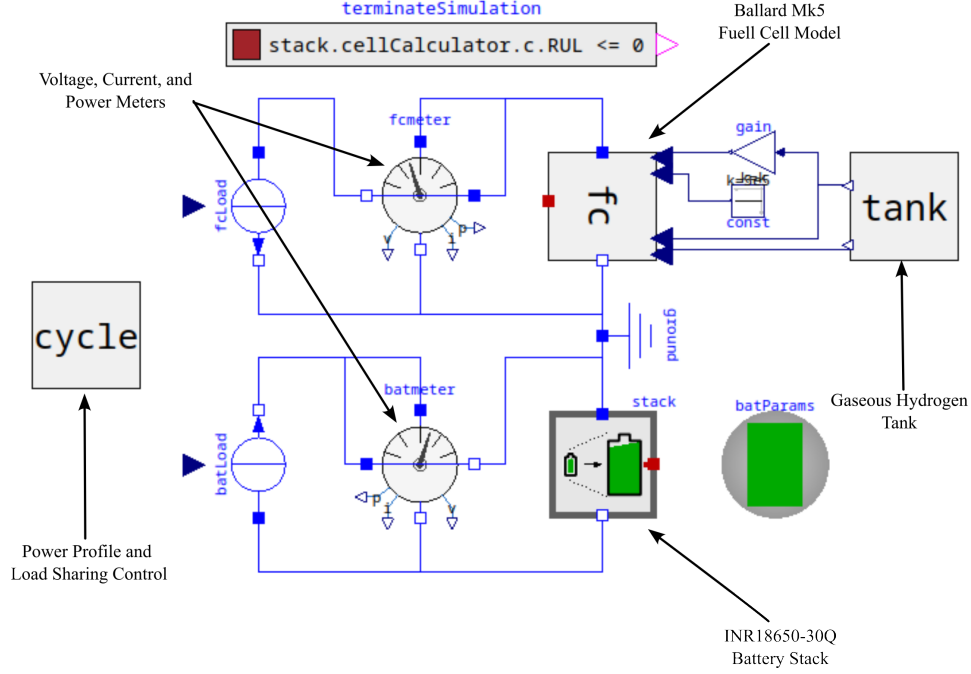


Fig. 4: The simplified power supply model in OpenModelica [8].

agent. A generic flight power profile was developed from publicly available flight data to characterize short distance flights around the southeastern United States. This basic template was then scaled up to match the power plant of the slightly larger Cessna 208. Combined, these blocks can simulate any number of flights of any length and provide a basis for the study of the battery stack's degradation over time.

A single hydrogen fuel cell can be modeled using a simple equivalent circuit model, as pictured in Fig. 5. Each component represents a specific phenomenon in the internal dynamics of the cell; the particulars of each are described in the subsections below. When combined in series, they describe the total voltage of a fuel cell [41, 42]

$$E_{fc} = NN_m(E_{OC} + V_{act} + V_{conc} + V_{ohm}), \quad (1)$$

where E_{OC} is the open circuit voltage, V_{act} is the activation losses, V_{conc} is the concentration losses, V_{ohm} is the ohmic losses, N_m is the number of fuel cell modules and N is the number of cells per module. Note that the direction of the pins for V_{act} , V_{conc} , V_{ohm} are opposite that of E_{OC} ; they therefore represent voltage *losses* rather than gains.

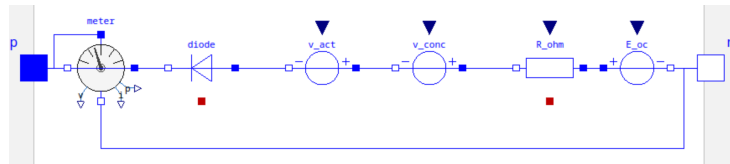
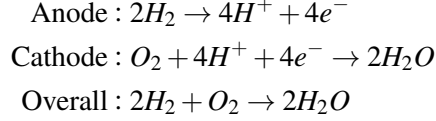


Fig. 5: Equivalent circuit model of a simple fuel cell.

In general, for any electrochemical cell, the open circuit voltage E_{OC} is defined by [41]

$$E_{OC} = \frac{\Delta G_{react}}{nF}, \quad (2)$$

where ΔG_{react} is the free energy change for the reaction, n is the number of electrons transferred in the reaction, and F is the Faraday constant. For a basic fuel cell, the global and half reactions are given in Eq. [41], which gives a value of $n = 2$. Generally speaking, air is provided as an oxidant rather than pure oxygen.



ΔG_{react} can further be split into two terms to adjust for reactions taking place at non-standard temperature and pressures, giving the final expression For a basic hydrogen fuel cell, this will be [41, 43]

$$E_{\text{OC}} = \frac{1}{nF} \left(\Delta G_{\text{react}}^{\circ} + RT \ln \left(\frac{p_{H_2} p_{O_2}^{0.5}}{p_{H_2O}} \right) \right). \quad (3)$$

The activation, concentration, and ohmic terms describe various internal dynamics of the cell, each of which results in a slight loss to the overall cell voltage. The activations losses are driven by the need to physically move electrons between the anode and cathode, while the concentration losses derive from discrepancies between the actual and ideal concentrations of reactants at each electrode. Finally, ohmic losses are caused by the internal resistance of the cell. These terms are all given by [41–43]:

$$V_{\text{act}} = a \ln i, \quad (4a)$$

$$V_{\text{conc}} = m e^{bi}, \quad (4b)$$

$$V_{\text{ohm}} = r i. \quad (4c)$$

The overvoltages given by Eq. 4 require a total of five parameters to fully characterize the cell: a , m , b , r , and the active area of the cell A . A set of empirical relations developed for a Ballard MK5 cell (active area 232 cm²) is used to characterize these [42] as a function of temperature.

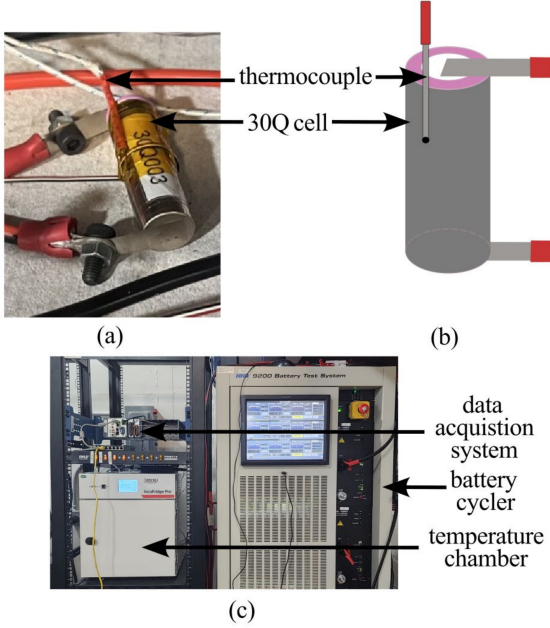
Table 1: Parameterization of a Ballard MK5 Fuel Cell.

parameter	expression
Activation loss coefficient, a (V)	$4.01 \times 10^{-2} - 1.40 \times 10^{-4} T$
Concentration loss coefficient, m (V)	$3.3 \times 10^{-3} - 8.2 \times 10^{-5} T$
Concentration loss power coefficient, b (cm ² ·mA ⁻¹)	8×10^{-3}
Cell resistance, r (kΩ·cm ²)	$4.77 \times 10^{-4} - 3.32 \times 10^{-6} T$

The battery stacks are based on the INR 18650-30Q, a graphite anode, LiCo₂ cathode cell manufactured by Samsung. Data for these cells was derived via performance testing at the University of South Carolina's Adaptive Real-Time Systems (ARTS) Lab. The open circuit voltage (OCV) is mapped to the cell's state of charge (SOC) by equation 5 [44], which governs the measured cell voltage as it is charged and discharged. The coefficients K_0 through K_3 were characterized via series of discharge tests. Tests were performed on an NHR 9200 battery system controlled by a programmable LabVIEW system (Fig. 6). The ambient temperature was maintained at 20 °C via a temperature control chamber. In each test, a single 18650-30Q was fully discharged at a constant current while recording the current, voltage, power, and temperature of

the cell. From this data, the cell SOC was estimated at each instant and curve fit to Eq. 5 using a least squares method. This process was repeated three times for a variety of discharge rates (0.1C, 1C, 2C, 3C, and 4C), resulting in five separate sets of coefficients, one for each discharge rate. Complete experimental results are available via a public repository [45]. This initial set of coefficients consistently overestimated the cell voltage by approximately 3-5%, depending on the discharge rate; each value of K_0 was reduced as appropriate, resulting in an excellent agreement with experimental results as shown in Fig. 7.

$$OCV = K_0 + K_1 SOC + K_2 \ln(SOC) + K_3 \ln(1 - SOC) \quad (5)$$



In addition, an empirical model from the literature was implemented to describe the capacity fade of the cell with repeated use. The model computes the cell's current capacity as a function of its nominal capacity (C_0), mean SOC (SOC_m), depth of discharge (DOD), and equivalent full cycles (EFC) and takes the form [46]

$$C = \frac{AC_0}{100} \left(\frac{EFC}{100} \right)^b, \quad (6)$$

where

$$A = 3.25 SOC_m \left(100 + 3.25 DOD - 2.25 DOD^2 \right). \quad (7)$$

The fuel cell is conceptualized as the primary power supply, providing full load during cruise, while the battery supplements the fuel cell's output during transients and acting as an emergency reserve. A simple load-sharing was applied to the power train as follows: in general, the aircraft is considered to pull from the fuel cell as its primary power supply. Thus, during the majority of the flight, the fuel cell receives the full load. However, during transient periods, the total load and fuel cell

Fig. 6: University of South Carolina Adaptive Real-Time Systems (ARTS) Laboratory, illustrating (a) and (b) the standard instrumentation for a 30Q cell and (c) the data acquisition, cycling, and temperature chamber.

load diverge. For example, during takeoff, the fuel cell slowly ramps up to cruise power, rather than swiftly up to maximum power and back. The difference required to complete takeoff is supplied by the battery. Similarly, during landing, the fuel cell ramps down more slowly than the total power demand of the aircraft, with the extra power produced being routed to charge the battery. To simulate the slight variations observed in the power profile of real flights, a noise generator was applied during cruise using a normal distribution with a standard deviation of 4.591 kW (approximately 1.37% of cruise power). When the signal generated is greater than the cruise power of 335 kW, the battery partially discharges to meet demand; when the signal is less than 335 kW, the fuel cell instead partially charges the battery. These charges and discharges, although small on an individual basis, add up over the course of a flight to increase the total equivalent cycles experienced by the stack and degrade it.

Results indicate battery lifespan to be sensitive to both the length of cruise time and the frequency and size of the fluctuations in the signal. In general, lifespan is inversely related to flight length; the longer the flight, the longer the battery stack will last. For example, in the ideal case (without fluctuations), stacks

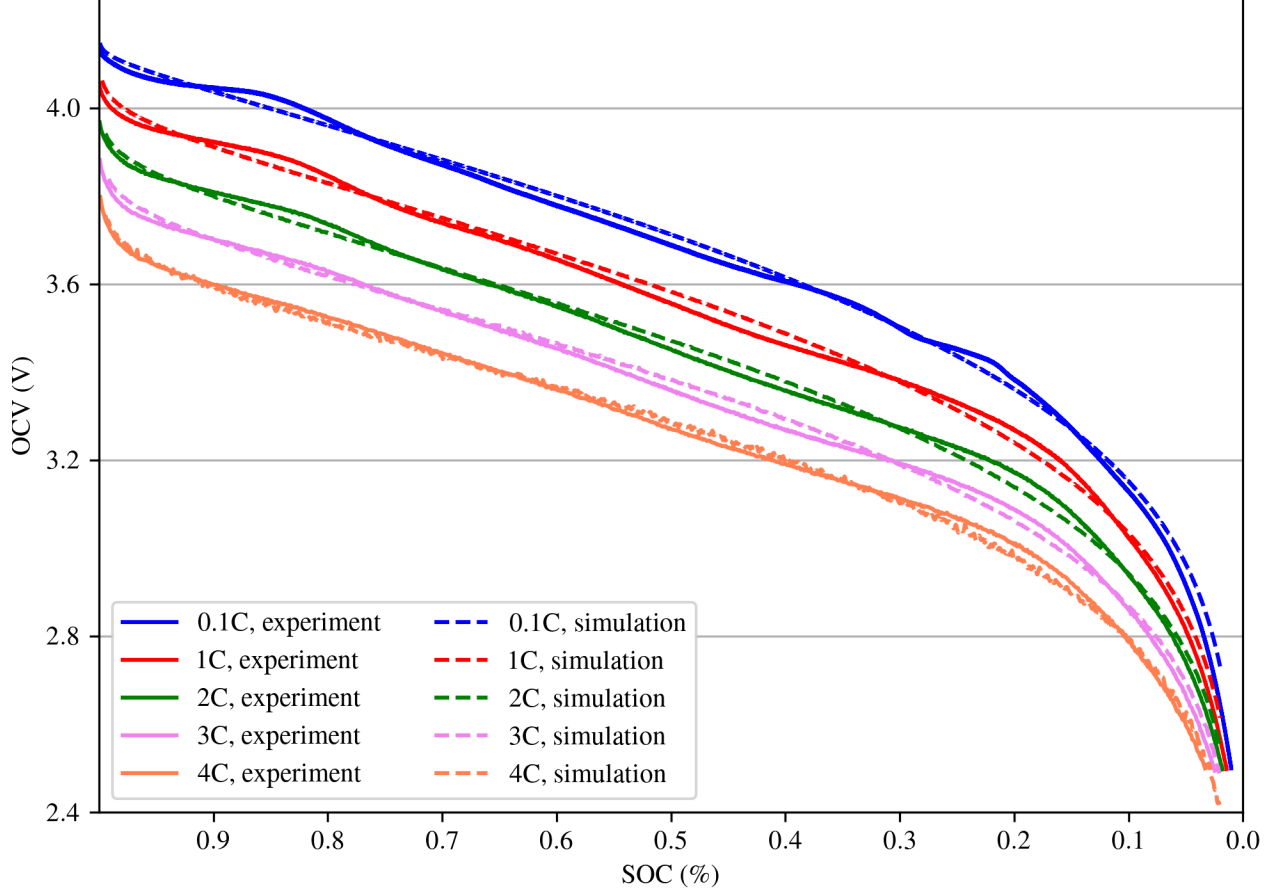


Fig. 7: 18650-30Q discharge curve at a variety of C rates.

subjected to short, 60-minute flights log 600 flight hours before replacement, compared to the just less than 1200 hours logged when engaging in 120-minute flights. In every flight, the deepest and most stressful discharge occurs during takeoff. The numerous small discharges occurring due to fluctuations in the load profile are minuscule, even in the aggregate, when compared with this initial discharge. Subsequently limiting an aircraft to repeated short-range flights means the battery stack will experience this stress more often than engaging with medium- and long-range flights.

Fluctuations in the load profile might introduced by air turbulence, mechanical failures, pilot error, or other irregularities. When these are accounted for, they can drastically reduce the overall lifespan of the stack. In the best case scenario (60-minute flights experiencing fluctuations 0.685% of cruise power), flight hours are reduced by roughly 10%, from 600 hours to 540 hours, while in the worst case (120-minute flights, 2.740% fluctuations) the life span shrinks by 34.8% from 1150 hours to 700 hours.

3 Intellectual Merit

The work in the its current state departs from convention in a number of ways. Firstly, the size and function of the battery has been scaled down greatly. In most studies, it is assumed that the stack is essentially interchangeable with the fuel cell. For example, in the distribution strategy devised by Li [37] the battery supplied between 20-30% of the load at any given time, peaking at approximately 35%. Meanwhile, in the drone flights performed by Lee [38] the battery frequently supplied over two-thirds of the power demand. Hence, the battery is almost continuously subjected to a sizable load, which in turn drastically increases the

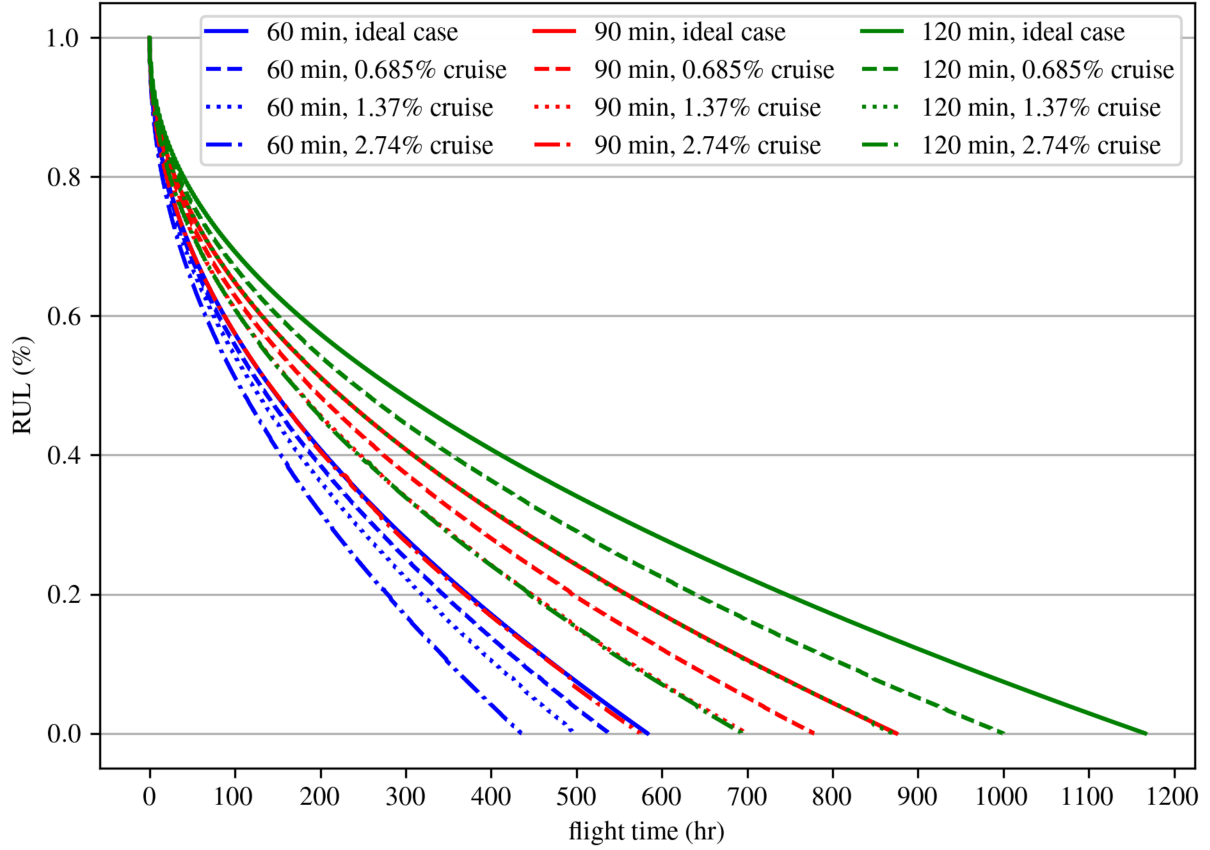


Fig. 8: Impact of fluctuations in the power profile on battery RUL.

required capacity. By focusing purely on takeoff and landing, the capacity and, crucially, the weight of the battery stack can be reduced. Furthermore, this approach allows each of the two power supplies to specialize and fill a role specific to their respective strengths.

Secondly, most studies are oriented towards optimizing energy usage, this being the primary goal of the ECMS strategy. However, this approach overlooks complexities and nuances in the economic model of airlines. Batteries represent a major maintenance bottleneck, and minimizing energy consumption does not necessarily equate to maximized battery lifespans. Subsequently, any cost savings attributed to energy usage may be overridden by capital costs associated with frequent battery replacement and revenue loss due to limited airframe availability. Adapting standard methods to also enhance battery lifespans maintains energy efficiency while additionally addressing logistical concerns.

Finally, to the author's knowledge no other study specifically investigates or accounts for small fluctuations in the load profile; generally speaking, in simulation space load profiles are smooth, and sometimes constant throughout cruise. The closest comparable efforts are studies involving physical hardware tests [38, 39], where such fluctuations are unavoidable. Developing a strategy that accounts for and manages these fluctuations addresses a specific gap in the existing literature.

4 Broader Impacts

Effective energy management systems are a key component in hybrid and all-electric power trains. By optimizing these systems for battery lifespan, the viability of electric aviation will be greatly improved. Furthermore, by developing models within an open-source framework, this project shall encourage academic and industry cooperation and facilitate future progress. The broader impacts include:

- **Environmental and economic benefits.** In addition to the environmental benefits attributable to reduced aircraft emissions, facilitating electric aviation is expected to yield economic returns. Widespread adoption is expected to lead to drastic reductions in the cost per kilowatt hour [47], with small-scale regional freight and passenger services especially benefitting [48]. Further, an EMS specifically designed to enhance battery life stands to minimize maintenance costs and maximize airframe availability.
- **Open sharing and promulgation.** At present, the overwhelming majority of EMS studies are dependent on commercial multi-physics programs. In contrast, all component models and controllers will be built upon and expand the open-source physics packages provided by OpenModelica. Furthermore, all code for these components will be made available via a public repository, as will all simulation scenarios and results. This approach ensures complete technical transparency and provides an open platform for academic collaboration and subsequent development.

5 Research Objectives

This work aims to design an energy management for a fuel cell/battery electric aircraft optimized to enhance battery lifespan via the following steps:

- **Objective 1: Complete the hybrid electric aircraft toolbox.** Expanding upon the base packages provided by OpenModelica, complete the physics models described in section 2.3 for the components of an aircraft power train, and, subsequently, design and program a set of load sharing controllers.
- **Objective 2: Demonstrate and fine-tune controllers on hardware.** Build a laboratory scale power train and demonstrate effective load sharing. Following successful demonstration, fine-tune controller to optimize the physical battery stack's useful life.
- **Objective 3: Explore the simulation space.** Identify key parameters impacting battery useful life. Quantify the effect of a range of values for these parameters on power train performance and economics.

6 Research Plan

6.1 Overview

The below tasks and subtasks are designed to complete each of the stated objectives. Task 1 completes an open-source Modelica library for the aircraft powertrain, including fuel cell, battery stack, fuel supplies, and thermal management systems and constructs and laboratory-scale testbed. Task 2 develops and calibrates a set of energy management controllers and validates their performance via hardware demonstration. Task 3 completes a sensitivity analysis and parametric study of the simulation space and constructs a surrogate model for aid in aircraft design.

6.2 Task 1: Develop multiphysics models and an experimental testbed framework

Develop an integrated simulation and experimental framework for studying fuel cell-battery hybrid aircraft by combining multidisciplinary modeling approaches with laboratory-scale validation. This task will establish a modular computational environment that connects multiphysics models of the aircraft power train, battery electrical and degradation models, fuel cell performance models, and mission-profile simulations within an open-source framework. By providing a flexible platform for evaluating different aircraft configurations, mission scenarios, and energy management strategies, this framework will support the development and assessment of advanced hybrid-electric aircraft concepts.

All models developed within the framework will be validated using data obtained from a laboratory-scale fuel cell-battery hybrid testbed to ensure accurate representation of system behavior under realistic operating conditions. Experimental validation will focus on evaluating model fidelity across nominal operation, dynamic load variations, and transient events representative of aircraft missions. The resulting simulation and testing capability will provide a comprehensive foundation for the analysis, optimization, and demonstration of advanced energy management systems for hybrid-electric aircraft.

Subtask 1.1: Develop modular multiphysics power train library

Develop a modular OpenModelica-based

library for each of the major aircraft power train components, including the fuel cell and battery stacks, hydrogen storage and supply, a thermal management loop, air supply, and power electronics. Each model shall be implemented so as to be easily reconfigured for alternate architectures, mission profiles, and load-sharing strategies. Ultracapacitor models will also be included to allow for additional power supply configurations.

Figure 9 shows an early version of a completed power train model with many of the final components. In addition to the fuel cell and battery models described above, the figure includes a liquid hydrogen storage tank, a hydrogen preheat interfacing with the cooling loop, compressor models for the hydrogen and air supply systems, and a heat exchanger. Additional components, such as the throttle valve, cooling loop PI, and coolant pump, are imported from the standard OpenModelica library. Not pictured are a time-averaged DC/DC booster and inverter, which are required to complete the electrical side of the power train.

Subtask 1.2: Integrate battery degradation and mission-profile modeling

Integrate physics-informed battery degradation models and realistic flight power profiles to quantify the impact on battery state of charge, depth of discharge, and remaining useful life. Degradation models will be based upon forms from the literature; in addition to the simple four-parameter model used above, which is suitable for the typical one voltage plateau cell, alternative models for more complex chemistries shall be made available to allow users further latitude in their design. Manufacturer data will be used for initial benchmarking, with the ARTS Laboratory providing additional data.

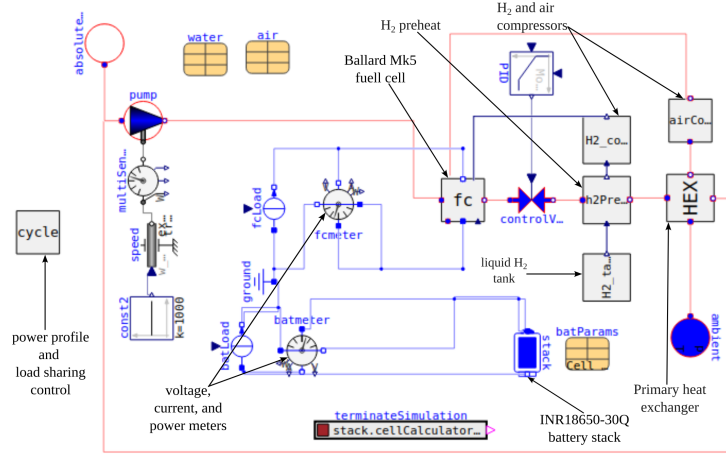


Fig. 9: Preliminary full power train model, including thermal loop for temperature control, liquid hydrogen storage, preheat, and supply, and air supply.

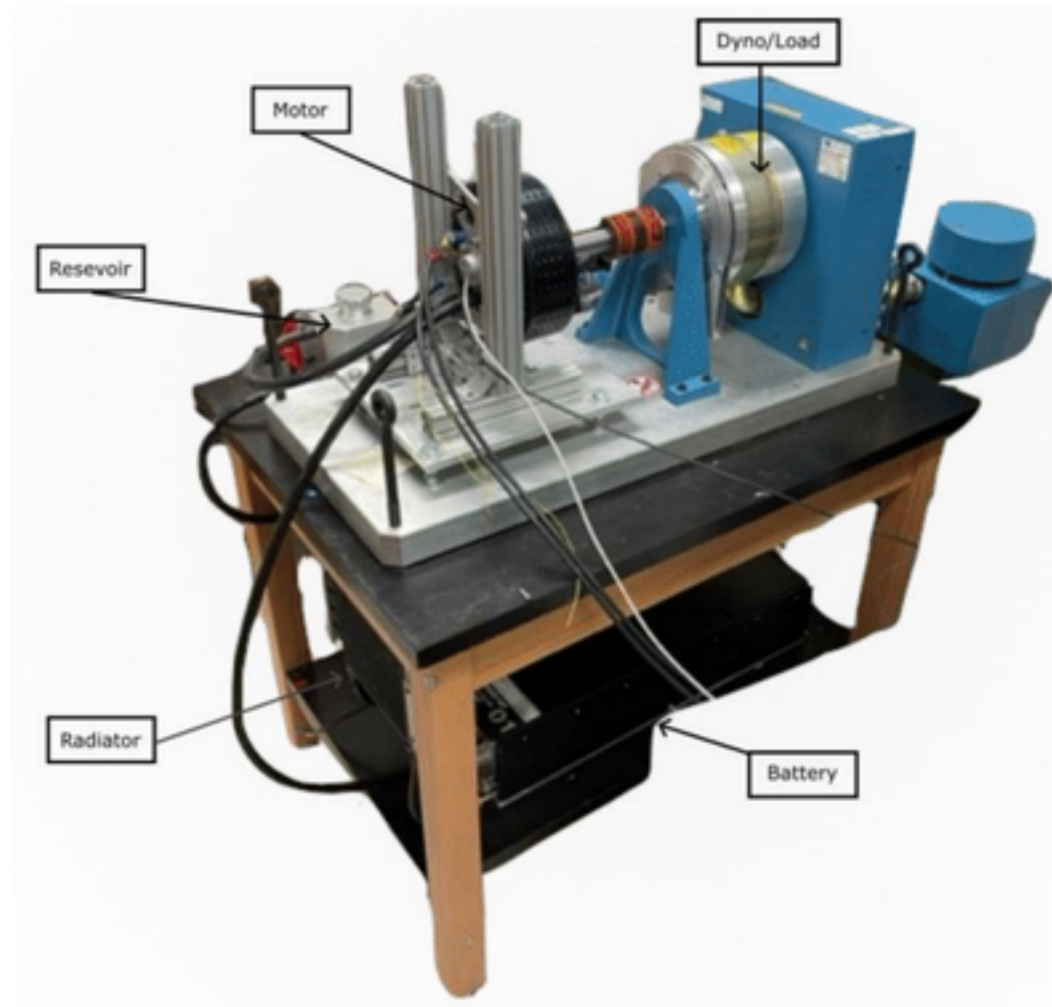


Fig. 10: Currently existing powertrain testbed at USC, including motor, damper, batter, and thermal loop. Modified testbed will incorporate a DC power supply to simulate fuel cell and a load sharing controller.

Initially, a simple flight profile consisting of taxi periods, takeoff, cruise, and landing shall be implemented, providing a baseline for performance. This done, flight irregularities, such as in-flight ascents and descents and random load fluctuations representing system noise, shall be introduced. Finally, emergency situations, such as aborted take-offs and landings or loss of one power supply will be studied. Combined, these modes provide a thorough spectrum of possible operating conditions. The completed framework is then used to identify specific conditions and parameters that accelerate battery aging and motivate controller requirements.

Subtask 1.3: Construct and validate a laboratory-scale hybrid powertrain testbed

Construct a laboratory-scale hybrid power train testbed to emulate a fuel cell/battery aircraft architecture. For this purpose, a pre-existing testbed (Figure 10) shall be repurposed and modified. As currently constituted, the testbed consists of small stack of INR 18650-30Q battery cells and an EMRAX 268 AC motor, a small thermal loop. The motor provides the testbed load and is equipped with a braking damper a controllers which may be used to modulate the load. The parallel/series configuration of the 30Qs may be reconfigured as desired to appropriately scale-down the battery stack present in the aircraft. Further, a programmable DC power supply (or equivalent emulator) will be added to act as a substitute for the fuel cell stack.

6.3 Task 2: Develop Autonomous Energy Management and Controller Optimization Methods

Develop autonomous energy management strategies that allocate power between the fuel cell and battery while preserving battery life, satisfying mission demand, and maintaining safe operation. This task will move beyond manual controller fine-tuning by studying autonomous calibration and optimization methods. Upon completion, this task shall provide a variety of simple and advanced load sharing strategies, as well as methods for calibrating the controllers for task-specific scenarios.

Subtask 2.1 Develop baseline and advanced EMS controllers

Develop autonomous energy management strategies that prioritize battery health while satisfying mission power demands and maintaining safe system operation. This research will investigate control architectures that intelligently allocate power between the battery and fuel cell to minimize battery degradation caused by high current transients, excessive depth of discharge, prolonged operation at extreme states of charge, and unfavorable thermal conditions. Rather than relying on static or manually tuned control rules, the proposed work will explore adaptive decision-making frameworks that leverage battery state estimation, degradation models, and mission awareness to dynamically balance system load.

This task will move beyond manual controller fine-tuning by investigating autonomous calibration and optimization methods that can systematically identify controller parameters for different applications and operating scenarios. Research will include the development and evaluation of both simple rule-based load-sharing strategies and more advanced optimization- and model-based approaches that improve system efficiency and robustness. Upon completion, this task shall deliver a suite of load-sharing algorithms along with automated controller calibration methods that can be tailored to task-specific mission requirements.

Subtask 2.2 Investigate autonomous controller calibration methods

Investigate autonomous methods for tuning controller parameters to improve battery remaining useful life while meeting load demand and observing fuel cell operating constraints. This research will focus on reducing the reliance on manual controller calibration by developing automated techniques that optimize controller performance across a range of operating conditions and mission profiles. Particular emphasis will be placed on identifying parameter sets that minimize battery degradation by limiting high-current events, avoiding excessive depth of discharge, and maintaining favorable operating conditions, while ensuring that the hybrid power system consistently satisfies mission power requirements and operates within established safety limits.

Candidate approaches may include optimization-based calibration, adaptive membership-function adjustment for fuzzy controllers, Bayesian optimization, and other data-driven methods that leverage high-fidelity simulation and experimental feedback to iteratively improve controller performance. The proposed research will evaluate the effectiveness, robustness, and computational efficiency of these techniques under representative mission scenarios and varying operating environments. The finalized methodologies will reduce controller development effort, improve repeatability across applications, and enable health-aware energy management strategies that maximize battery remaining useful life without compromising system performance or fuel cell reliability.

Subtask 2.3 Validate autonomous EMS performance on the hardware testbed

Validate the selected controller strategies using the laboratory-scale testbed developed in Task 1. Experimental testing will be conducted to assess controller performance under representative operating conditions while verifying that the proposed energy management approaches can reliably satisfy system power demands and maintain stable operation. Validation activities will examine the ability of each controller to regulate power sharing between the battery and fuel cell surrogate and respond effectively to dynamic changes in system loading. Test results will be compared against simulation predictions to quantify model fidelity and identify opportunities for controller refinement.

Controller performance will be evaluated under nominal flight profiles, realistic load fluctuations, and abnormal operating scenarios representative of challenging mission conditions, including aborted takeoff, rapid descent, and irregular transient power demand. Performance metrics will include power tracking accuracy, transient response, battery state-of-charge management, battery health indicators, fuel cell operating stability, overall system efficiency, and robustness to disturbances. The experimental results will establish the relative strengths and limitations of each control strategy and provide the data necessary to assess their readiness for deployment in hybrid-electric aircraft power systems while guiding future improvements to autonomous energy management algorithms.

6.4 Task 3: Develop Kriging-Based Surrogate Models for Hybrid Electric Aircraft Performance and Economics

Develop computationally efficient surrogate models that map aircraft design and operating parameters to key performance, degradation, and economic outcomes for hybrid electric aircraft. This task will leverage data generated from the physics-based simulations and test bed results produced in Tasks 1 and 2 to develop surrogate models capable of predicting system interdependencies across a broad design space. These models will enable efficient exploration of hybrid-electric aircraft design trade offs and support accelerated evaluation of emerging propulsion concepts.

Subtask 3.1 Investigate sensitivity of battery life and cost to design parameters

Investigate the impact of key design and operating parameters on battery life, hydrogen consumption, power train mass, replacement interval, and mission feasibility. This research will quantify how variations in controller settings, mission profiles, and operational constraints influence the overall performance and economic viability of hybrid-electric aircraft. Because these design variables are highly coupled, changes intended to improve one performance metric may produce unintended consequences in others. The proposed work will therefore examine these interactions using sensitivity analysis techniques to better understand the complex trade spaces that govern hybrid power system design.

The sensitivity study will identify the parameters that most strongly influence hybrid-electric aircraft performance and economics, providing insight into the relative importance of battery configuration and specifications, energy management strategies, overall mass, and mission requirements. Statistical and computational analysis methods will be used to rank influential variables, characterize nonlinear interactions, and identify combinations of parameters that maximize system performance while minimizing life cycle costs. The resulting analyses will support informed design decisions and guide further development priorities.

Subtask 3.2 Develop parametric design-space simulations

Develop a comprehensive parametric simulation campaign using the validated system models and energy management controllers developed in Tasks 1 and 2. The study will systematically evaluate the performance of hybrid-electric aircraft across a broad design space by varying aircraft- and mission-level parameters that influence energy consumption, component utilization, and overall system economics. High-fidelity simulations will be used to capture the interactions between the battery, fuel cell, and mission profile, enabling a consistent assessment of performance under representative operational scenarios.

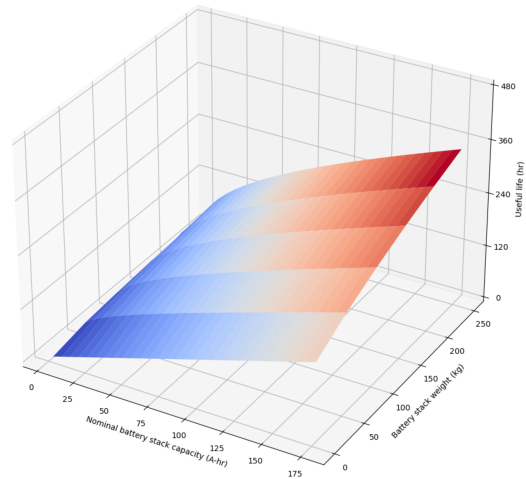


Fig. 11: Conceptual example of the power train surrogate model, mapping stack capacity and weight onto useful life.

The simulation campaign shall iterate through aircraft-relevant design variables, including battery size, fuel cell size, flight range, cargo payload, cruise power demand, load fluctuation magnitude, depth-of-discharge limits, recharge strategy, hydrogen and electricity prices, battery and fuel cell replacement cost, and controller parameters. Advanced design-of-experiments and parametric analysis techniques will be employed to efficiently explore the multidimensional design space and identify trends, sensitivities, and interactions among the selected variables. The resulting dataset will establish quantitative relationships between design choices and key performance metrics.

Subtask 3.3 Develop Kriging-based surrogate models for aircraft-level performance

Develop Kriging-based surrogate models that map the parametric simulation results to aircraft-level outcomes, including battery useful life, hydrogen consumption, maintenance burden, life cycle cost, and aircraft availability. The surrogate models will provide computationally efficient approximations of the validated high-fidelity simulations, enabling rapid evaluation of large design spaces and complex interactions among aircraft configuration, mission requirements, energy management strategies, and operating conditions. By capturing nonlinear relationships between input variables and system-level performance metrics, these models will support accelerated design exploration while reducing the computational expense associated with repeated high-fidelity simulations.

The developed surrogate models will be used to identify optimal design and control strategies that minimize life cycle cost while preserving mission performance and operational reliability for hybrid-electric aircraft. Optimization frameworks will leverage the surrogate models to evaluate trade offs between component sizing, energy management decisions, battery degradation, hydrogen utilization, maintenance requirements, and aircraft availability. The resulting analysis will provide quantitative guidance for selecting architectures and control strategies that balance economic performance with mission capability, enabling more informed development of efficient, reliable, and sustainable hybrid-electric aircraft systems.

7 Project Management Plan

The research plan shall proceed allowing the following approximate timeline:

- **August 2026:** Completion of physical component models (Task 1.1). Begin construction of testbed (Task 1.3) with aid of undergraduate students.
- **September 2026:** Complete degradation models (Task 1.2) and basic EMS controller models (Task 2.1) and obtain initial results (Task 1.2).
- **October 2026:** Complete testbed construction (Task 1.3).
- **November and December 2026:** Calibrate controllers (Task 2.2) and validate results on testbed (Task 2.3).
- **January 2026:** Perform sensitivity analysis and parametric study (Task 3.1 and 3.2).
- **February 2027:** Build surrogate model (Task 3.3) and submit final results for defense and publication.

Upon completion, the project is expected to yield multiple conference publications, at least one journal publication, and a doctoral dissertation. In addition, all component and controller models shall be published as part of a publicly available repository [49] for use with the OpenModelica platform. Finally, the testbed built as part of Task 2 shall remain as a basis for future studies.

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References Cited

- [1] B. Graver, K. Zhang, and D. Rutherford, “Co2 emissions from commercial aviation, 2018,” Tech. Rep.
- [2] “Consolidated statement of continuing icao policies and practices related to environmental protection - general provisions, noise, and local air quality,” International Civil Aviation Organization, Tech. Rep. A42-WP/24, October 2025.
- [3] X.-G. Yang, T. Liu, S. Ge, E. Rountree, and C.-Y. Wang, “Challenges and key requirements of batteries for electric vertical takeoff and landing aircraft,” *Joule*, vol. 5, no. 7, pp. 1644–1659, Jul. 2021, [Online]. Available: <https://doi.org/10.1016/j.joule.2021.05.001>
- [4] K. Adebajo, G. Anthony, J. Peskar, A. R. Downey, Y. Chen, and C. Hu, “Impact of charging rate and state of charge on electric aircraft battery degradation using a multi-domain model with realistic southeastern u.s. flight paths,” in *AIAA SCITECH 2025 Forum*. American Institute of Aeronautics and Astronautics, Jan. 2025, [Online]. Available: <https://doi.org/10.2514/6.2025-2705>
- [5] C. Y. Justin, A. P. Payan, S. I. Briceno, B. J. German, and D. N. Mavris, “Power optimized battery swap and recharge strategies for electric aircraft operations,” *Transportation Research Part C: Emerging Technologies*, vol. 115, p. 102605, Jun. 2020, [Online]. Available: <https://doi.org/10.1016/j.trc.2020.02.027>
- [6] P. Fragiaco, M. Genovese, F. Piraino, O. Corigliano, and G. De Lorenzo, “Hydrogen-fuel cell hybrid powertrain: Conceptual layouts and current applications,” *Machines*, vol. 10, no. 12, pp. 1121–1140, July 2022, [Online]. Available: <https://doi.org/10.3390/machines10121121>
- [7] J. Bauman and M. Kazerani, “A comparative study of fuel-cell-battery, fuel-cell-ultracapacitor, and fuel-cell-battery-ultracapacitor vehicles,” *IEEE Transactions on Vehicular Technology*, vol. 57, no. 2, pp. 760–769, March 2008, [Online]. Available: <https://doi.org/10.1109/TVT.2007.906379>
- [8] N. Cooper, A. Mahboub Rad, K. Adebajo, A. R. J. Downey, and J. Zhang, “Physics-based prediction of remaining useful life of lithium-ion batteries in fuel cell-battery hybrid aircraft using an open-source modelica framework,” ser. ASME International Mechanical Engineering Congress and Exposition, vol. Volume 9: Micro- and Nano-Systems Engineering and Packaging; Safety Engineering, Risk and Reliability Analysis; Special Symposium on Additive Manufacturing on Benchmark Test Series; Special Symposium on Power; Research Posters, 11 2025, p. V009T15A017. [Online]. Available: <https://doi.org/10.1115/IMECE2025-168114>
- [9] M. Andersson and E. Inberg, “Life cycle assessment of a regional hybrid electric aeroplane,” Master’s thesis, Chalmers University of Technology, Gothenburg, Sweden, 2024.
- [10] T. Benjamin, R. Borup, N. Garland, C. Gittleman, B. Habibzadeh, S. Hirano, D. Ho, G. Kleen, and e. a. Kopasz, John, “Usdrive fuel cell technical team roadmap,” 2017.
- [11] OpenModelica Association, *OpenModelica User’s Guide*, 2023. [Online]. Available: <https://openmodelica.org/doc/OpenModelicaUsersGuide/1.21/>
- [12] T. Zeng, C. Zhang, M. Hu, Y. Chen, C. Yuan, J. Chen, and A. Zhou, “Modelling and predicting energy consumption of a range extender fuel cell hybrid vehicle,” *Energy*, vol. 165, pp. 187–197, 2018, [Online]. Available: <https://doi.org/https://doi.org/10.1016/j.energy.2018.09.086>

- [13] J.-J. Hwang, Y.-J. Chen, and J.-K. Kuo, "The study on the power management system in a fuel cell hybrid vehicle," *International Journal of Hydrogen Energy*, vol. 5, pp. 4476–4489, 2012, [Online]. Available: <https://doi.org/https://doi.org/10.1016/j.ijhydene.2011.11.127>
- [14] T. C. Do, H. V. A. Truong, H. V. Dao, M. H. Cong, X. D. To, T. D. Dang, and K. K. Ahn, "Energy management strategy of a pem fuel cell excavator with a supercapacitor/battery hybrid power source," *Energies*, vol. 12, no. 22, pp. 24 362–4386, November 2019, [Online]. Available: <https://doi.org/10.3390/en12224362>
- [15] J. Jia, G. Wang, Y. T. Cham, Y. Wang, and M. Han, "Electrical characteristic study of a hybrid pemfc and ultracapacitor system," *IEEE Transactions on Industrial Electronics*, vol. 57, no. 6, pp. 1945–1953, June 2010, [Online]. Available: <https://doi.org/10.1109/TIE.2009.2022066>
- [16] "Electra: Opening a new era of aviation," <https://www.electra.aero/>, accessed: 2026-04-17.
- [17] "Hybrid-electric aircraft: Ampaire," <https://www.ampaire.com/>, accessed: 2026-04-17.
- [18] "Zeroavia makes aviation history, flying world's largest aircraft powered with a hydrogen-electric engine," <https://zeroavia.com/do228-first-flight/>, accessed: 2026-04-17.
- [19] A. R. Gnadt, R. L. Speth, J. S. Sabnis, and S. R. Barrett, "Technical and environmental assessment of all-electric 180-passenger commercial aircraft," *Progress in Aerospace Sciences*, vol. 105, 2019, [Online]. Available: <https://doi.org/https://doi.org/10.1016/j.paerosci.2018.11.002>
- [20] "Velis electro - pipistrel," <https://www.pipistrel-aircraft.com/products/velis-electro/>, accessed: 2026-04-17.
- [21] N. K. Borer, S. C. Geuther, B. L. Litherland, and L. W. Kohlman, "Design and performance of a hybrid-electric fuel cell flight demonstration concept," in *2018 Aviation Technology, Integration, and Operations Conference*, 2018, p. 3357.
- [22] S. Clarke, M. Redifer, K. Papathakis, A. Samuel, and T. Foster, "X-57 power and command system design," in *2017 IEEE Transportation Electrification Conference and Expo (ITEC)*. IEEE, 2017, pp. 393–400.
- [23] P. Garcia, J. P. Torreglosa, L. M. Fernandez, and F. Juado, "Control strategies for high-power electric vehicles powered by hydrogen fuel cell, battery, and supercapacitor," *Expert Systems with Applications*, vol. 40, pp. 4791–4804, 2013, [Online]. Available: <https://doi.org/https://doi.org/10.1016/j.eswa.2013.02.028>
- [24] M. Hannan, F. Azidin, and A. Mohamed, "Multi-sources model and control algorithm of an energy management system for light electric vehicles," *Energy Conversion and Management*, vol. 62, pp. 123–130, 2012, [Online]. Available: <https://doi.org/https://doi.org/10.1016/j.enconman.2012.04.001>
- [25] M. Ouyang, L. Xu, J. Li, L. Lu, D. Gao, and Q. Xie, "Performance comparison of two fuel cell hybrid buses with different powertrain and energy management strategies," *Journal of Power Sources*, vol. 163, pp. 467–479, 2006, [Online]. Available: <https://doi.org/https://doi.org/10.1016/j.jpowsour.2006.09.033>
- [26] J. Torreglosa, F. Juado, P. Garcia, and L. Fernandez, "Hybrid fuel cell and battery tramway control based on an equivalent consumption minimization strategy," *Control Engineering Practice*, vol. 19, pp. 1182–1194, 2011, [Online]. Available: <https://doi.org/https://doi.org/10.1016/j.conengprac.2011.06.008>

- [27] L. Xu, J. Li, J. Hua, X. Li, and M. Ouyang, "Optimal vehicle control strategy of a fuel cell/battery hybrid city bus," *International Journal of Hydrogen Energy*, vol. 34, pp. 7323–7333, 2009, [Online]. Available: <https://doi.org/https://doi.org/10.1016/j.ijhydene.2009.06.021>
- [28] P. Rodatz, G. Paganelli, A. Sciarretta, and L. Guzzella, "Optimal power management of an experimental fuel cell/supercapacitor-powered hybrid vehicle," *Control Engineering Practice*, vol. 13, pp. 41–53, 2005, [Online]. Available: <https://doi.org/https://doi.org/10.1016/j.conengprac.2003.12.016>
- [29] D. Xiao and Q. Wang, "The research of energy management strategy for fuel cell hybrid vehicle," in *2012 International Conference on Industrial Control and Electronics Engineering*, 2012, pp. 931–934, [Online]. Available: <https://doi.org/https://doi.org/10.1109/ICICEE.2012.247>
- [30] C.-Y. Li and G.-P. Liu, "Optimal fuzzy power control and management of fuel cell/battery hybrid vehicles," *Journal of Power Sciences*, vol. 192, pp. 525–533, 2009, [Online]. Available: <https://doi.org/https://doi.org/10.1016/j.jpowsour.2009.03.007>
- [31] Y. Eren, O. Erdinc, H. Gorgun, M. Uzunoglu, and B. Vural, "A fuzzy logic based supervisory controller for an fc/uc hybrid vehicular power system," *International Journal of Hydrogen Energy*, vol. 34, pp. 8681–8694, 2009, [Online]. Available: <https://doi.org/https://doi.org/10.1016/j.ijhydene.2009.08.033>
- [32] A. Arce, A. J. del Real, and C. Bordons, "Mpc for battery/fuel cell hybrid vehicles including fuel cell dynamics and battery performance improvement," *Journal of Process Control*, vol. 19, pp. 1289–1304, 2009, [Online]. Available: <https://doi.org/https://doi.org/10.1016/j.jprocont.2009.03.004>
- [33] Q. Chen, L. Gao, R. A. Dougal, and S. Quan, "Multiple model predictive control for a hybrid proton exchange membrane fuel cell system," *Journal of Power Sciences*, vol. 191, pp. 473–482, 2009, [Online]. Available: <https://doi.org/https://doi.org/10.1016/j.jpowsour.2009.02.034>
- [34] T. M. Vu, R. Moezzi, J. Cyrus, J. Hlava, and M. Petru, "Parallel hybrid electric vehicle modelling and model predictive control," *Applied Sciences*, vol. 11, no. 22, 2021, [Online]. Available: <https://doi.org/https://doi.org/10.3390/app112210668>
- [35] A. R. E. Wise, S. Sumsurooah, S. Yeoh, S. Bozhko, and O. Hebala, "Extended comparative analysis of aircraft energy management strategies for improvements in energy expenditure, hydrogen savings, and battery lifecycle assessment," *Aerospace*, vol. 13, 2026, [Online]. Available: <https://doi.org/https://doi.org/10.3390/aerospace13030251>
- [36] S. Njoya Motapon, L.-A. Dessaint, and K. Al-Addad, "A comparative study of energy management schemes for a fuel-cell hybrid emergency power system of more-electric aircraft," *IEEE Transactions on Industrial Electronics*, vol. 61, no. 3, pp. 1320–1334, 2014, [Online]. Available: <https://doi.org/https://doi.org/10.1109/TIE.2013.2257152>
- [37] S. Li, C. Gu, P. Zhao, and S. Cheng, "A novel hybrid propulsion system configuration and power distribution strategy for light electric aircraft," *Energy Conversion and Management*, vol. 238, 2021, [Online]. Available: <https://doi.org/https://doi.org/10.1016/j.enconman.2021.114171>
- [38] B. Lee, S. Kwon, P. Park, and K. Kim, "Active power management system for an unmanned aerial vehicle powered by solar cells, a fuel cell, and batteries," *IEEE Transaction on Aerospace and Electronic Systems*, vol. 50, 2014, [Online]. Available: <https://doi.org/https://doi.org/10.1109/TAES.2014.130468>
- [39] X. Zhang, L. Liu, Y. Dai, and T. Lu, "Experimental investigation on the online fuzzy energy management of hybrid fuel cell/battery power system for uavs," *International Journal of Hydrogen Energy*, vol. 43, 2018, [Online]. Available: <https://doi.org/https://doi.org/10.1016/j.ijhydene.2018.04.075>

- [40] “CHEETA - Center for High-Efficiency Electrical Technologies for Aircraft,” <https://cheeta.illinois.edu/>, accessed: 2026-04-17.
- [41] J. Larminie and A. Dicks, *Fuel Cell Systems Explained*. John Wiley & Sons Ltd., 2003.
- [42] F. Laurencelle, R. Chahine, J. Hamelin, K. Agbossou, M. Fournier, T. K. Bose, and A. Laperrière, “Characterization of a ballard mk5-e proton exchange membrane fuel cell stack,” *Fuel Cells*, vol. 1, no. 1, pp. 66–71, 2001, [Online]. Available: [https://doi.org/https://doi.org/10.1002/1615-6854\(200105\)1:1<66::AID-FUCE66>3.0.CO;2-3](https://doi.org/https://doi.org/10.1002/1615-6854(200105)1:1<66::AID-FUCE66>3.0.CO;2-3)
- [43] S. N. M., O. Tremblay, and L.-A. Dessaint, “A generic fuel cell model for the simulation of fuel cell vehicles,” in *2009 IEEE Vehicle Power and Propulsion Conference*, 2009, pp. 1722–1729, [Online]. Available: <https://doi.org/10.1109/VPPC.2009.5289692>
- [44] C. Chin, Z. Gao, and C. Zhang, “Comprehensive electro-thermal model of 26650 lithium battery for discharge cycle under parametric and temperature variations,” *Journal of Energy Storage*, vol. 28, p. 101222, Apr. 2020, [Online]. Available: <https://doi.org/10.1016/j.est.2020.101222>
- [45] ARTS-Lab, “Dataset electrothermal deformation characterization for Samsung 30q cell,” GitHub, 2024. [Online]. Available: <https://github.com/ARTS-Laboratory/dataset-electrothermal-deformation-characterization-for-samsung-30Q-cell>
- [46] S. Saxena, C. Hendricks, and M. Pecht, “Cycle life testing and modeling of graphite/lico2 cells under different state of charge ranges,” *Journal of Power Sources*, vol. 327, pp. 394–400, 2016. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0378775316309247>
- [47] J. Hospodka, H. Bělovský, and S. Pleninger, “Assessment of all-electric general aviation aircraft,” *Energies*, vol. 13, 2020. [Online]. Available: <https://www.mdpi.com/1996-1073/13/23/6206>
- [48] A. Prapotnik Brdnik, R. Kamnik, M. Marksel, and S. Bozicnik, “Market and technological perspectives for the new generation of regional passenger aircraft,” *Energies*, vol. 12, 2019. [Online]. Available: <https://www.mdpi.com/1996-1073/12/10/1864>
- [49] ARTS-Lab, “Open Modelica modeling of battery fuel cell electric aircraft,” GitHub. [Online]. Available: <https://github.com/ARTS-Laboratory/Open-Modelica-modeling-of-battery-fuel-cell-electric-aircraft>